

E_Lecture4_W1

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Q1 [30 pts, 15 pts for each]

(1)

如果效用函數為 $h(Q_1^D, Q_2^D) = A \cdot Q_1^D \cdot Q_2^D$ ，原來的消費組合 $(Q_1^D, Q_2^D) = (20, 30)$ ，當 Q_1^D 增為 25 時， Q_2^D 是多少方會與 $(20, 30)$ 在同一無異曲線上？

Suppose that the utility function is $h(Q_1^D, Q_2^D) = A \cdot Q_1^D \cdot Q_2^D$, where Q_1^D and Q_2^D are the consumption volume for product 1 and 2, respectively.

If the original consumption bundle is $(Q_1^D, Q_2^D) = (20, 30)$ and the consumer would like to increase the Q_1^D from 20 to 25, how does he adjust Q_2^D so that he can keep his utility unchanged (i.e., let $(25, Q_2^D)$ and $(20, 30)$ be on the same indifference curve)?

$$Q_2^D \cdot 25A = 30 \cdot 20A$$

$$Q_2^D = 24$$

From $(20, 30)$ change to $(25, 24)$ while the utility is unchanged.

(2)

在前題中， A 值的大小對答案有何影響？

Would the value A in the utility function affect the answer of Q_2^D ?

No effect, unless $A = 0$, then the value of utility will become 0.

Q2	[30 pts, 15 pts for each] [In-class exercise 4-1] (1) 令 Q_1^D 表包子消費個數， Q_2^D 表香蕉消費根數。某人最喜歡的消費組合 $(Q_1^D, Q_2^D) = (3, 2)$，吃的太多或太少效用均不及消費 $(3, 2)$ 時的水準。無異曲線形狀將為何？ Q_1^D is the consumption volume of buns and Q_2^D is that of bananas. A consumer's optimal consumption bundle is $(Q_1^D, Q_2^D) = (3, 2)$. The utility level resulting from consuming more or less buns and bananas will be smaller than $(3, 2)$. The indifference curve will be: 呈現環狀的無異曲線，離中心點越遠，所帶來的效用越小，反之亦然。 (2) 若消費者對 (Q_1^D, Q_2^D) 的偏好如上所述，請問其無異曲線是否符合「負斜率」、「凸向原點」的特性？ Would the indifference curves have negative slope and be bowed inward? This case does not obey the property.
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Q3	[30 pts, 5 pts for each] Consider that a consumer named Hurley. His income is \$1200, and he consumes only fish and mango Prices: $P_F = \\$4$ per fish, $P_M = \\$1$ per mango. (A) If Hurley spends all his income on fish, how many fish does he buy? $Q_F = 1200/4 = 300$ (B) If Hurley spends all his income on mangos, how many mangos does he buy?
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$$Q_M = 1200/1 = 1200$$

(C)

If Hurley buys 100 fish, how many mangos can he buy?

$$Q_M = 1200 - (100 \cdot 4) = 800$$

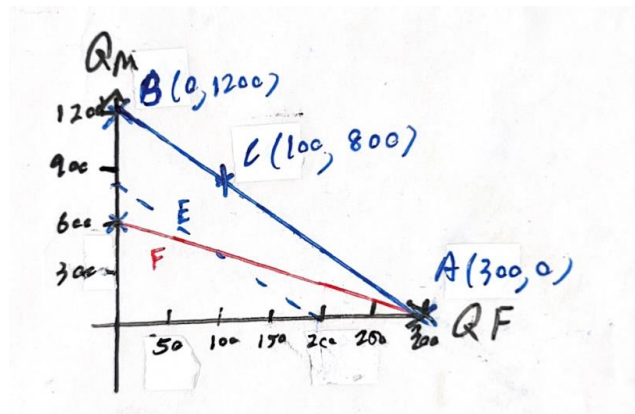
(D)

Plot each of the bundles from parts (a) to (c) on a graph that measures fish on the horizontal axis and mangos on the vertical; connect the dots

Show what happens to Hurley's budget constraint if

(E) His income falls to \$800.

(F) The price of mangos rises to $P_M = \$2$ per mango, given his income is \$1200.



Q4	[10 pts, 5 pts for each]
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(1)

Emilio buys pizza for \$10 and soda for \$2. He has income of \$100. His budget constraint will experience a **parallel outward shift** if which of the following events occur?

- The price of pizza falls to \$5, the price of soda falls to \$1, and his income falls to \$50.
- The price of pizza rises to \$20, the price of soda rises to \$4, and his income remains the same.
- The price of pizza falls to \$8, the price of soda falls to \$1, and his income rises to \$120.
- The price of pizza rises to \$20, the price of soda rises to \$4, and his income rises to \$400.

Answer: D

(2)

At any point on an indifference curve, the slope of the curve measures the consumer's

- a) income.**
- b) willingness to trade one good for the other.**
- c) perception of two goods as substitutes or complement.**
- d) elasticity of demand.**

Answer: B